

Design Project: ANALOG/DIGITAL AOM DRIVER CIRCUIT WITH AWG CONTROL AND PHOTODIODE TESTING

Course: ECE1895, University of Pittsburgh

Professor: Dr. Samuel Dickerson

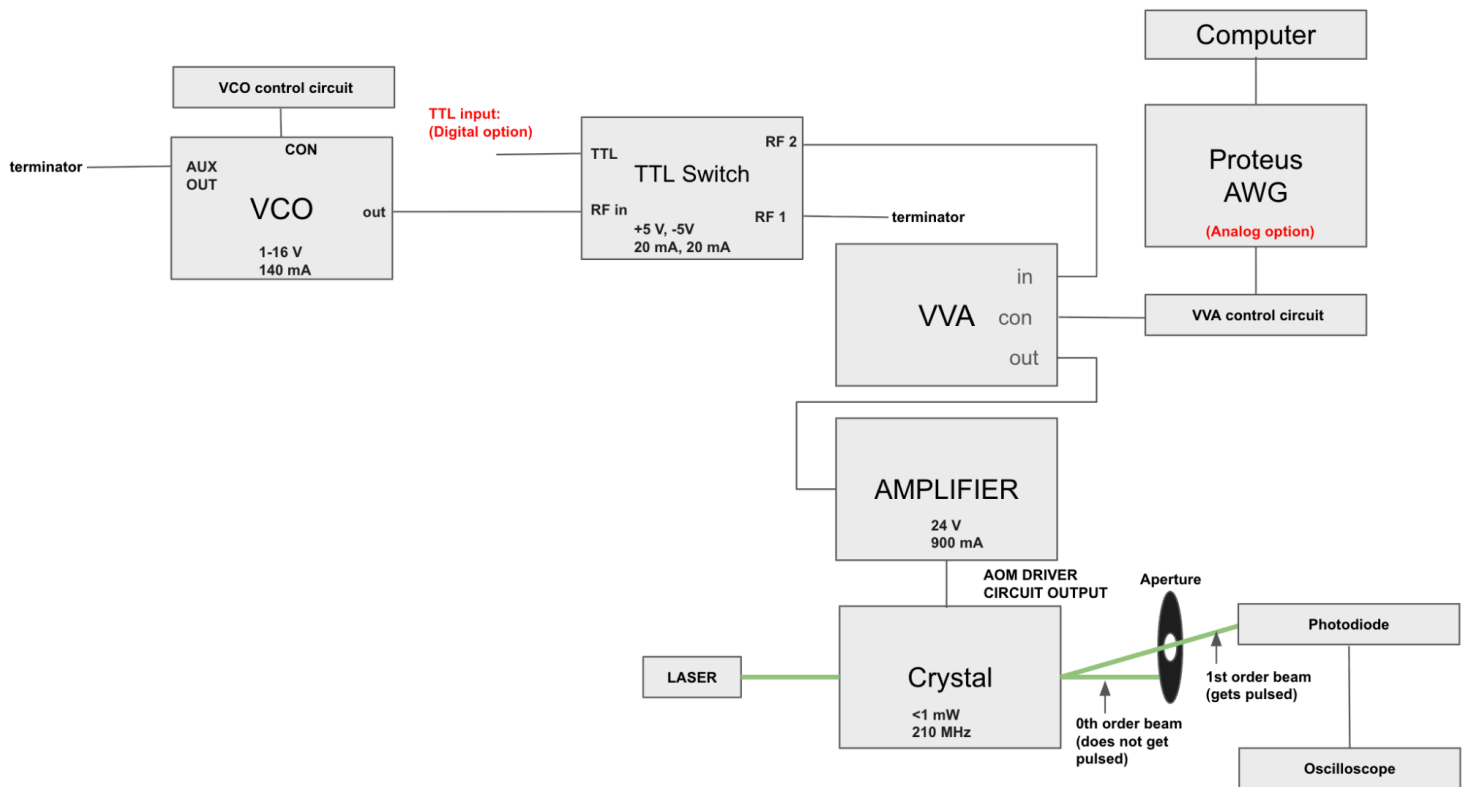
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Design Overview

- This report goes over how an acousto-optic modulator driver circuit is built. The idea is that the acousto-optic modulator (AOM) is a crystal that diffracts light. When a signal is applied to the crystal at the resonance frequency, the light diffraction can be controlled by amplitude modulation and therefore, the on and off times can be controlled by a TTL switch (digital option) and the amplitudes can be controlled by the voltage variable attenuator (VVA) (analog option). An arbitrary waveform generator (AWG) is then programmed to generate pulses that define the pulse shape of the wave. Initial design plan was to use a 24V supply and linear regulators. However, after testing, it was found that the components heated up and therefore, the idea switched to using a computer power supply and step up converters.

Figure 1: Block diagram of the complete AOM driver system with AWG control and photodiode testing

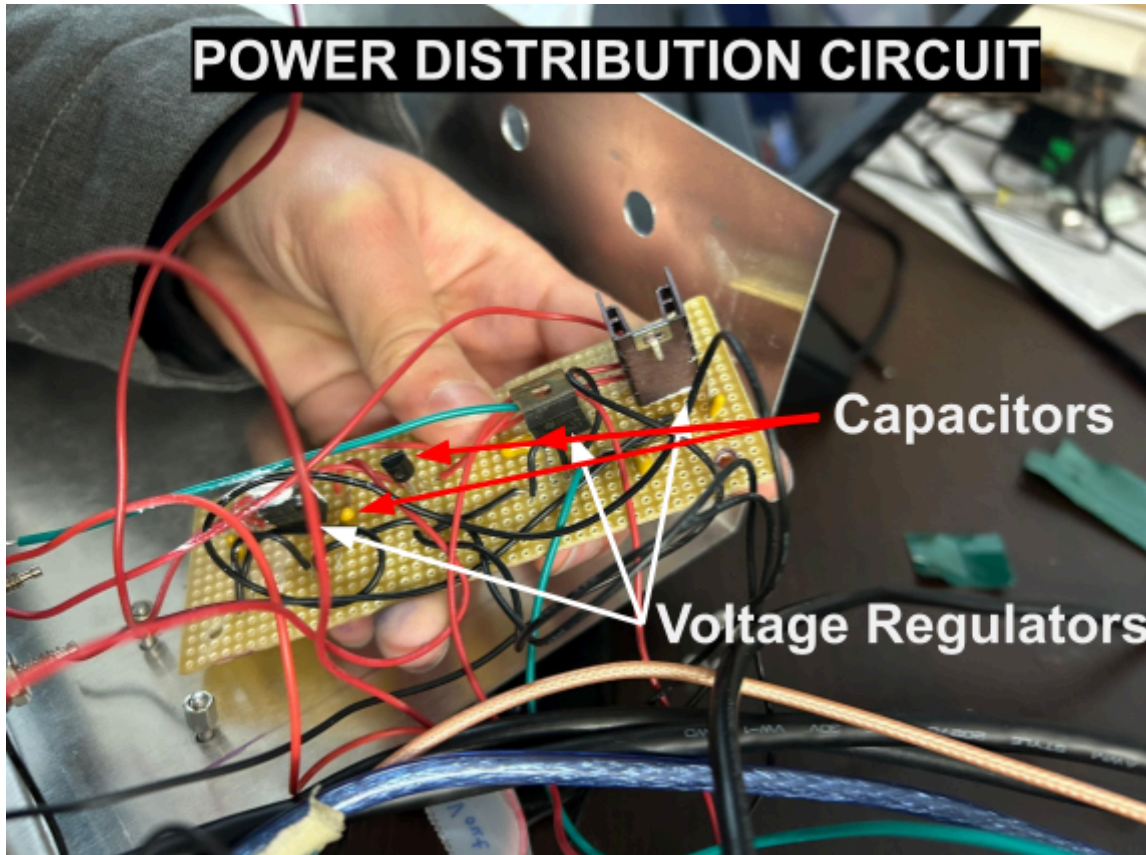
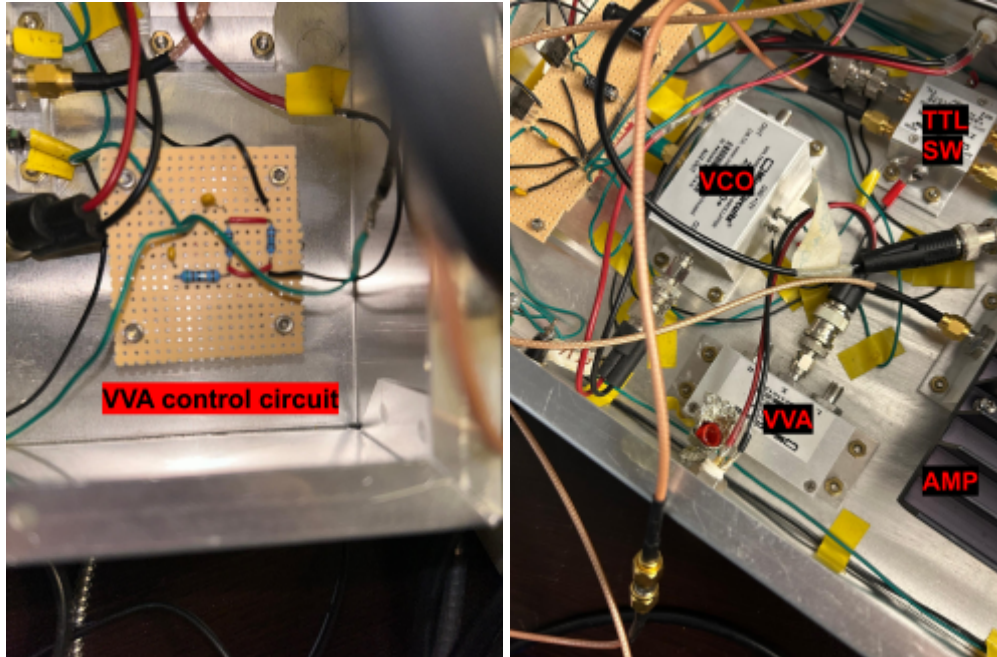


- The original design consists of a VCO, VVA, TTL switch, and an Amp. The power distribution was made of linear voltage regulators and a lab-bench power supply. The original system was non-functional. To ensure component safety, the original board was not powered. The bulk components were extracted and power distribution was built using a computer power supply and step up converters. Variable voltage control for the VCO was added to make this circuit adjustable for different crystals with different frequencies. The PCB was built and the circuit was tested by programming the Proteus AWG to control the amplitudes. A laser is used for testing. It was sent through the AOM and the first order beam (the controlled beam) is sent to a photodiode, which outputs the AOM diffraction results.
- This circuit uses old components from a broken circuit, builds analog/digital hybrid AOM driver, adds frequency control, implements AWG control, introduces PCB for compact design, and uses a photodiode for robust testing. While the original circuit was a fixed-frequency analog-only driver, this project expands it into a tunable hybrid analog/digital system with software control, representing a complete redesign and functional upgrade suitable for a 6-week design cycle.

Preliminary Design Verification

The old circuit components are shown in figure 2 and the circuit was not powered for safety. Components were extracted one by one. The circuit schematic is shown in Figure 3, and RF devices were tested separately based on their specs.

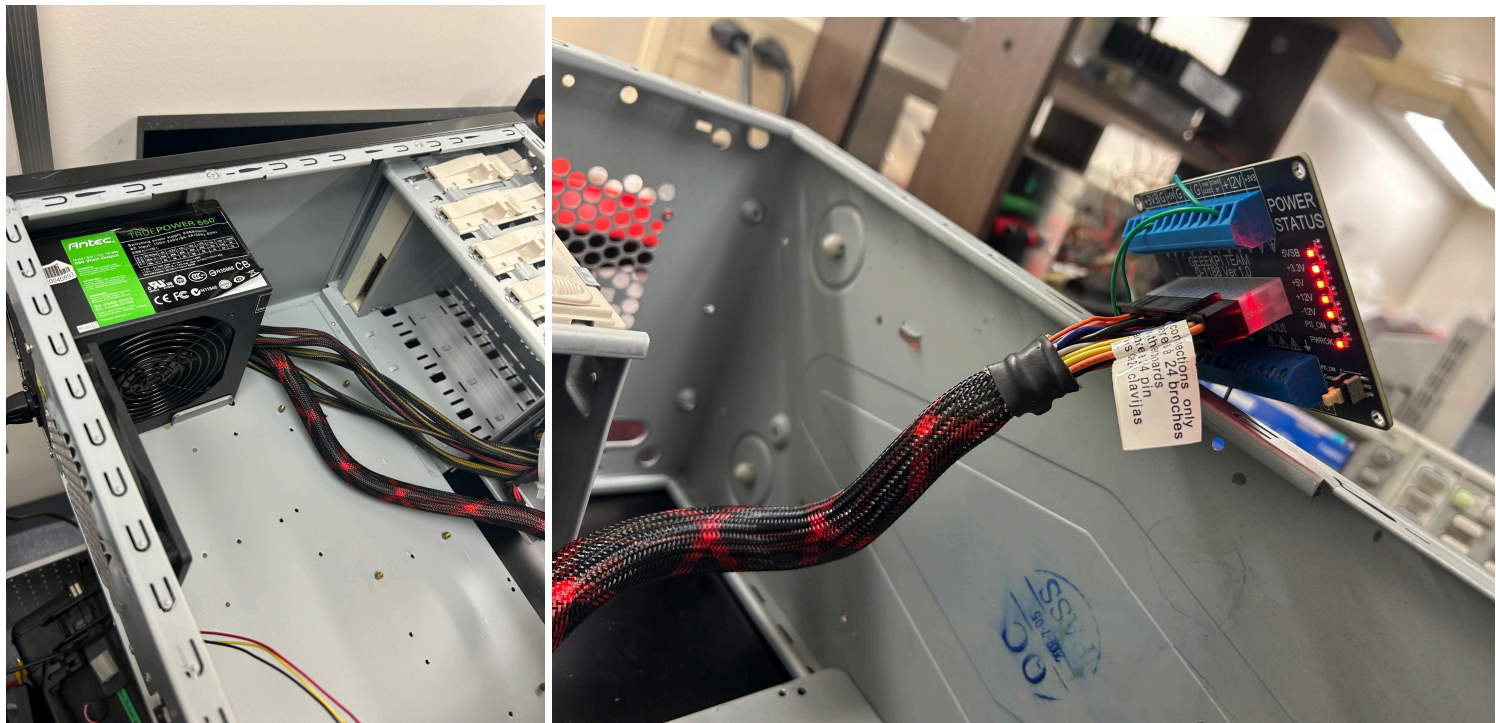
Figure 2:Old circuit



The diagram illustrates a computer power supply system for an optical experiment. Key components and connections include:

- Power Input:** A 16V source is connected to a 2317 voltage regulator.
- Regulation and Control:** The 2317 regulator is controlled by a 20kΩ potentiometer. Its output is connected to a VCO power and control block.
- Voltage Conversion:** A 12V to 16V step-up transformer is used to convert the 12V input to 16V. Another 12V to 24V step-up transformer is used to convert the 12V output of the VCO block to 24V.
- Measurement:** A V/A meter with an attenuator and a 50Ω load is connected to the -5V line. The meter's output is connected to an A/M P block.
- Optical Path:** The output of the A/M P block is connected to an aperture, which is then connected to a photodiode. The photodiode's output is connected to an AOM crystal, which is finally connected to a laser.
- Other Components:** A TTL switch is connected to the 12V line. A 7905 negative voltage regulator is connected to the 12V line to provide a -5V output. A 2.2μF capacitor is connected to the 12V input of the 7905. A 1μF capacitor is connected to the -5V output of the 7905.

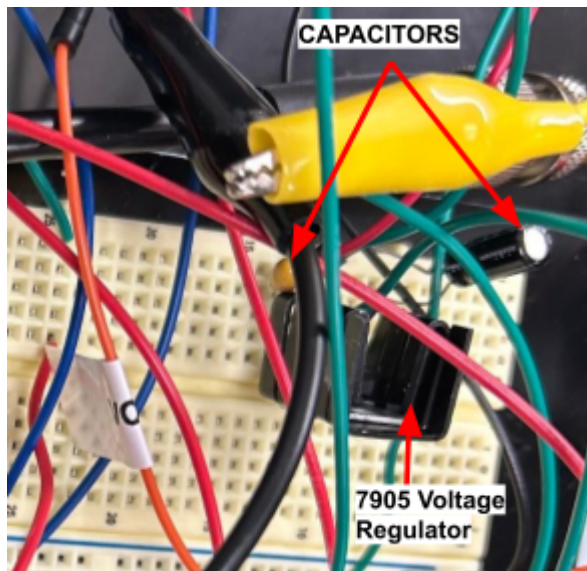
Figure 4: Computer power supply with ATX breakout board.



This provided access to -12 V, 12V, and 5V. However, the circuit also needed -5V, 24 V, and 16 V. So a 7905 voltage regulator circuit was used to convert -12 V to -5V needed to supply the TTL switch (Figure 5)

Two step up converters were used: one for the amplifier (12 V to 24 V) and one for the VCO control (12 V to 16V).

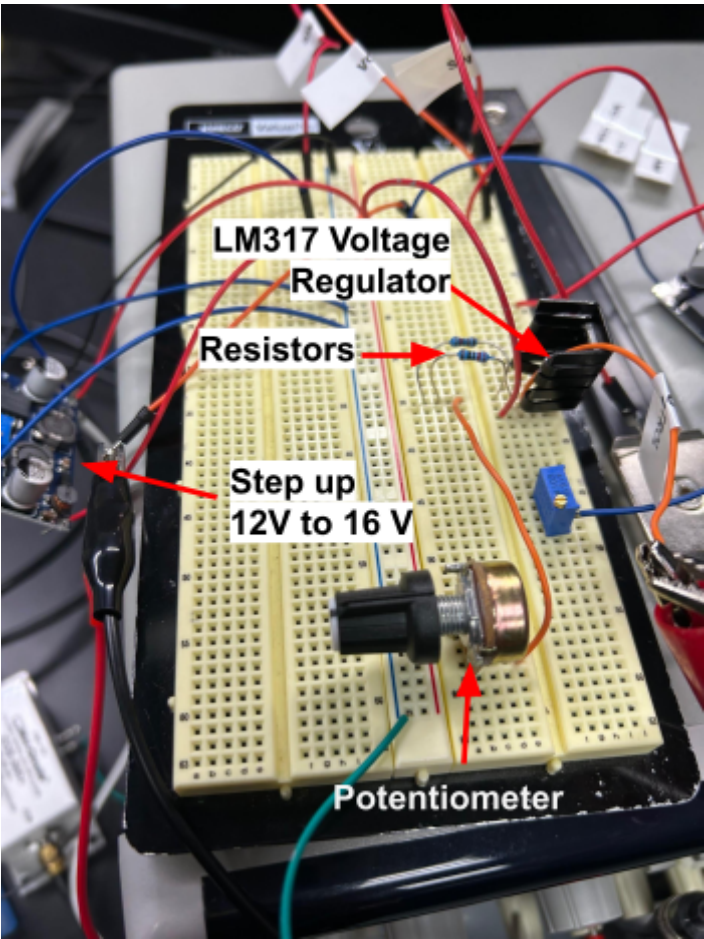
Figure 5: a. Function generator setting (TTL) b. 7905 voltage regulator circuit to convert -12 V to -5V for TTL switch



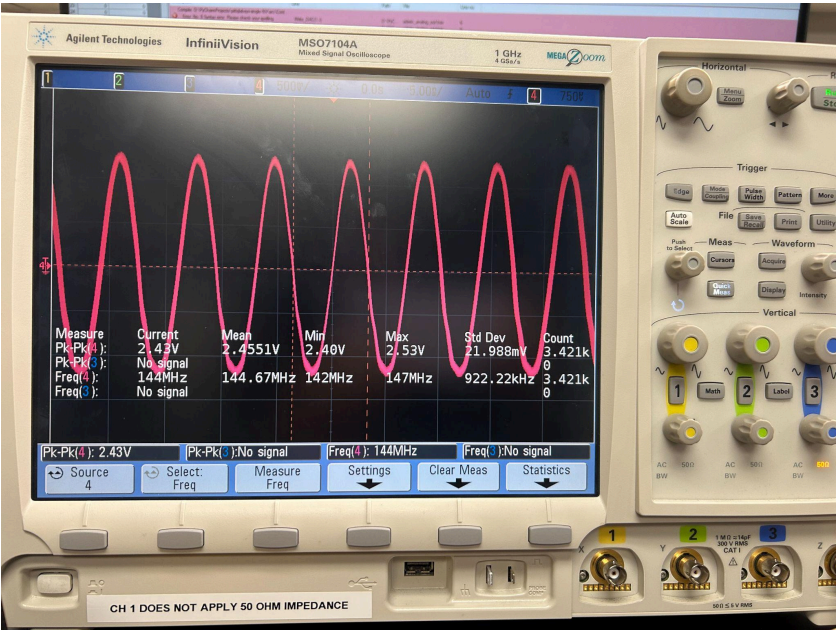
The 16V goes to LM317 voltage regulator circuit and varies the output from 1 V to 16 V using a 20 k Ω potentiometer to vary the VCO frequency from about 150 MHz to 300 MHz (Figure 6).

Figure 6: VCO Control Breadboard prototype a. Circuit, b. Min frequency, c. max frequency

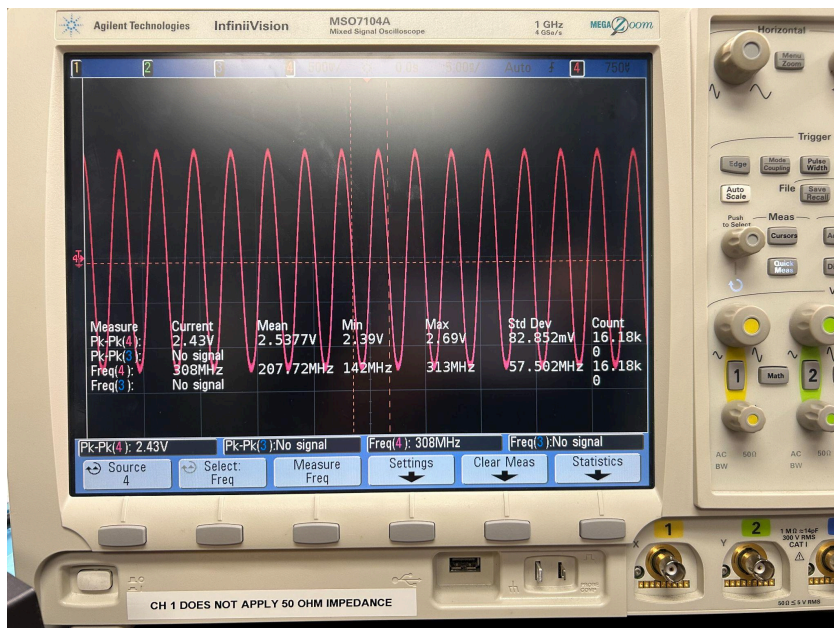
a.



b.

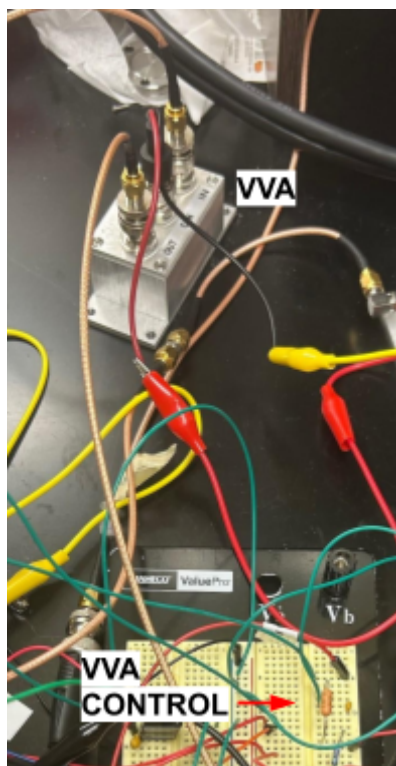


c.



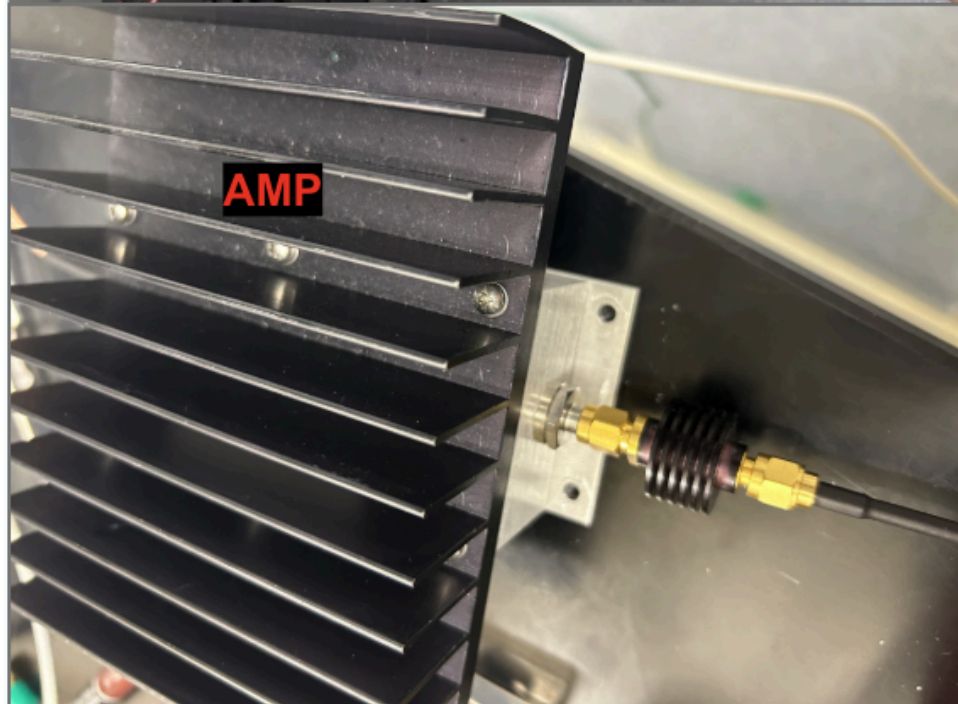
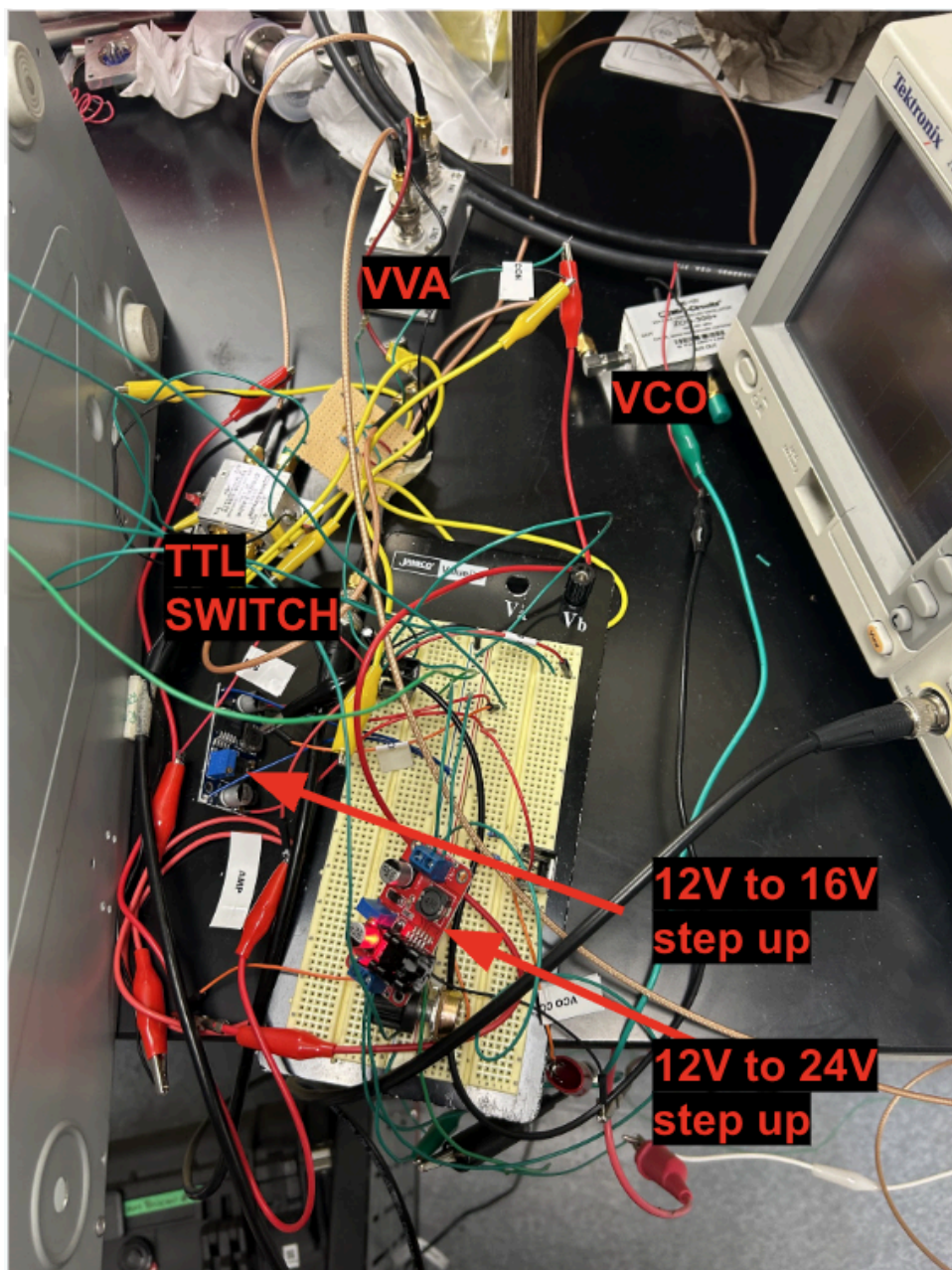
Then, a VVA control circuit was built following the suggested design in reference [1] (Figure 7)

Figure 7: VVA control circuit



Once independent breadboard circuits worked, the full circuit was assembled (figure 8)

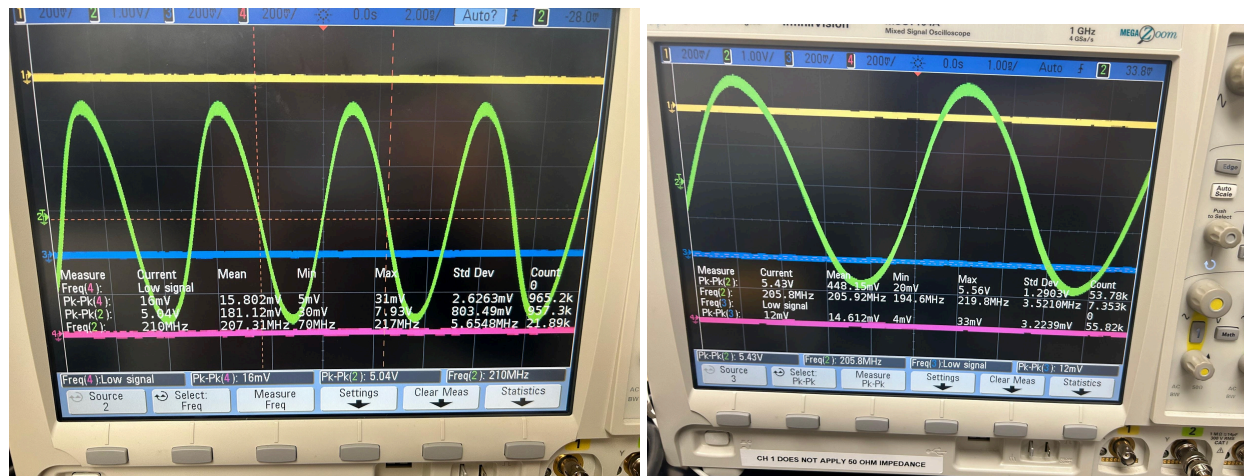
Figure 8: Full Breadboard Prototype Circuit:



The AOM requires $<1\text{ W}$ into 50Ω . Using $P = (V_{\text{rms}})^2/R$, the maximum allowable V_{rms} is $\sqrt{1 \times 50} \approx 7.07\text{ V}$, which corresponds to $2\sqrt{2}V_{\text{rms}} = 20\text{ Vpp}$ for a sine wave. The voltage was measured on an oscilloscope at 50Ω impedance to ensure that this limit was not exceeded. Our oscilloscope only handles 7V so a fixed attenuator was added before the amplifier.

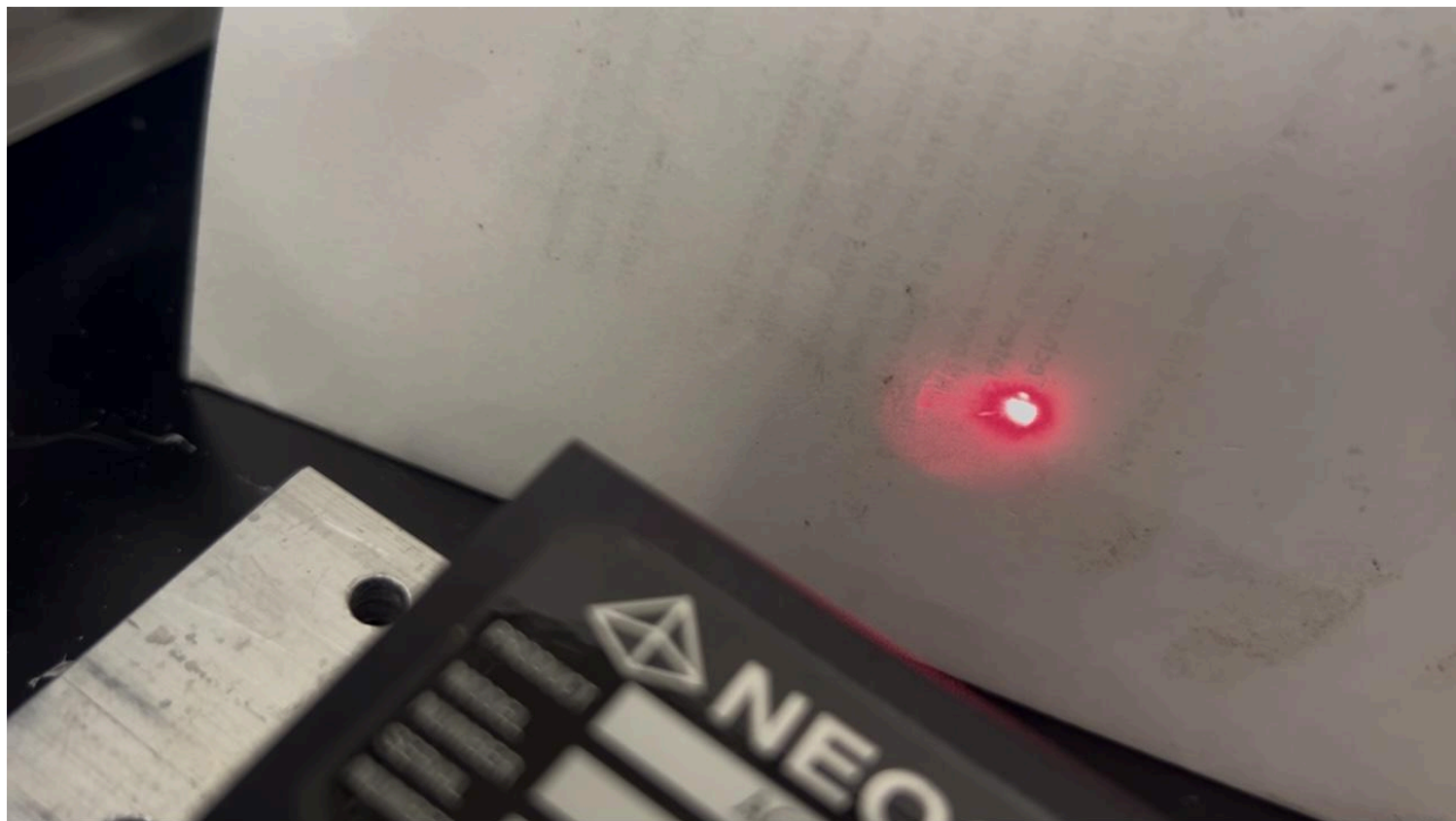
Both analog and digital versions of the circuit were tested (Figure 9)

Figure 9: 5Vpp full circuit output on an oscilloscope at 210 MHz (crystal frequency) [a 20 dB attenuator was used to protect the oscilloscope]:



Finally, the circuit was connected to an AOM and a laser pointer was used to test if there is diffraction. (Figure 10)

Figure 10: Optical diffraction pattern from AOM. The zero-order beam (undiffracted) and first-order diffracted beam (right) are visible when RF signal is applied at 210 MHz.



Challenges and solutions:

- The amplifier step up converter did not provide enough power; it was replaced with a higher power step up converter from JB superstore.
- A 24 V power supply was initially considered to be used with voltage regulators for power distribution, but the components overheated even with heatsinks, so the design switched to using a computer power supply along with step up converters. The only regulators used are LM317 for the VCO control voltage and 7905 to regulate -12V to -5 V for the TTL switch.

Feasibility was confirmed by successfully demonstrating each subcircuit on a breadboard with measured output comparable to design targets before PCB layout began.

Design Implementation

The final system operates as follows: The AWG generates a modulation control signal, which is sent to the VVA (analog mode). Since our AWG does not generate TTL, we use a function generator to test the digital option and we send the signal to the TTL switch. The VCO produces an RF carrier, which we tune using the 20 k Ω potentiometer to optimize it at the crystal's resonance frequency, which is then modulated (VVA or TTL switch or both), amplified, and sent to the AOM. A photodiode verifies optical diffraction.

Power Distribution

- Computer PSU (12V, 5V, -12V) → step-up converters (24V for amp, 16V for VCO control) → 7905 (-5V for TTL switch)
- 7905 Linear regulator (-12V → -5V) for switch supply

Hardware Control

- 16V to LM317 circuit: DC voltage tuned using 20k Ω potentiometer (VCO control)
- AWG signal sent through a simple control circuit provided in reference [1] (VVA Control)

Final Component List

RF chain

- TTL switch (Reference [4])
- VVA (Reference [5])
- Amp (Reference [2])
- VCO (Reference [3])

Test Equipment

- Proteus AWG 1284M (Reference [7])
- Keysight PXI Chassis (Reference [8])
- Function Generator
- Aperture
- NEOS15210 AOM (Reference [9]) (or any other AOM crystal with a resonance frequency from 150 MHz to 300 MHz)
- A laser
- Photodiode (Reference [10])
- Oscilloscope

Power

- Computer power supply (Reference [11])
- ATX pins
- 7905 (through hole) (Reference [6])
- 2 x heat sinks
- 2 x step up converters
- 1 μF (1206)
- 2.2 μF

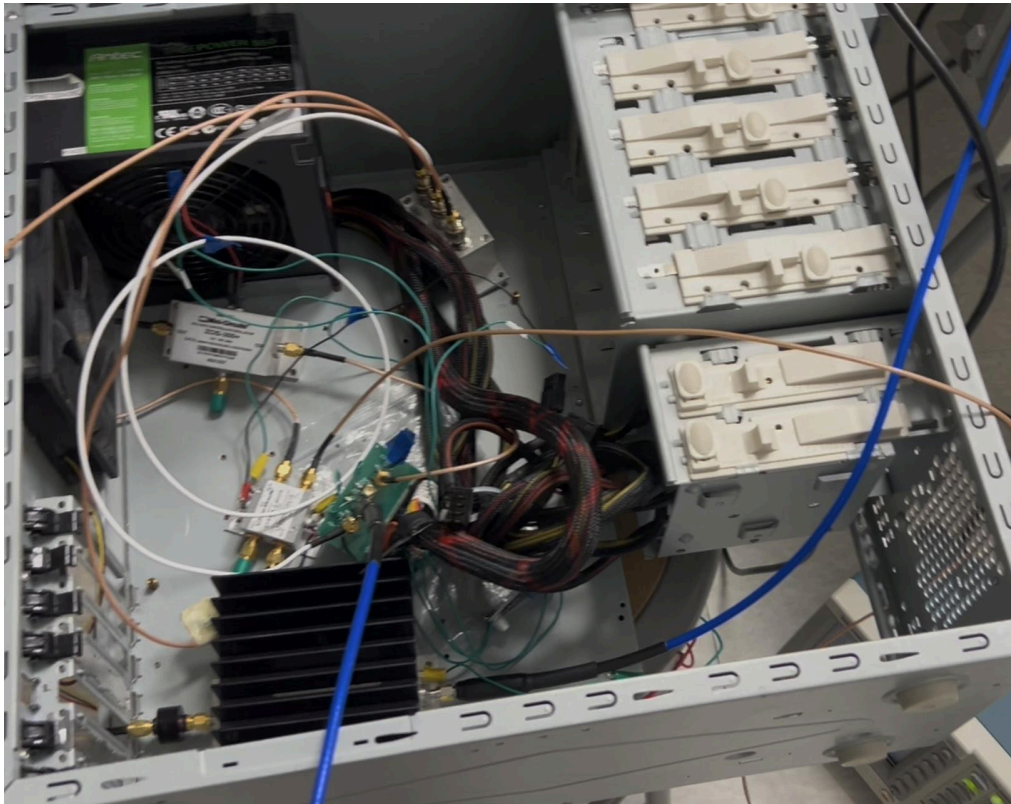
Control

- LM317 (through hole) (Reference [5])
- 220 Ω (0805)
- 1.2 k Ω (0805)
- 51 Ω (0805)
- 500 Ω (1206)
- 2 x 0.01 μF (1206)
- 20 k Ω potentiometer (through hole)

Final Enclosure

The circuit and components were placed in a computer power supply which is metal for shielding the components. It also has a spot for the power supply, and a fan for cooling. (Figure 11)

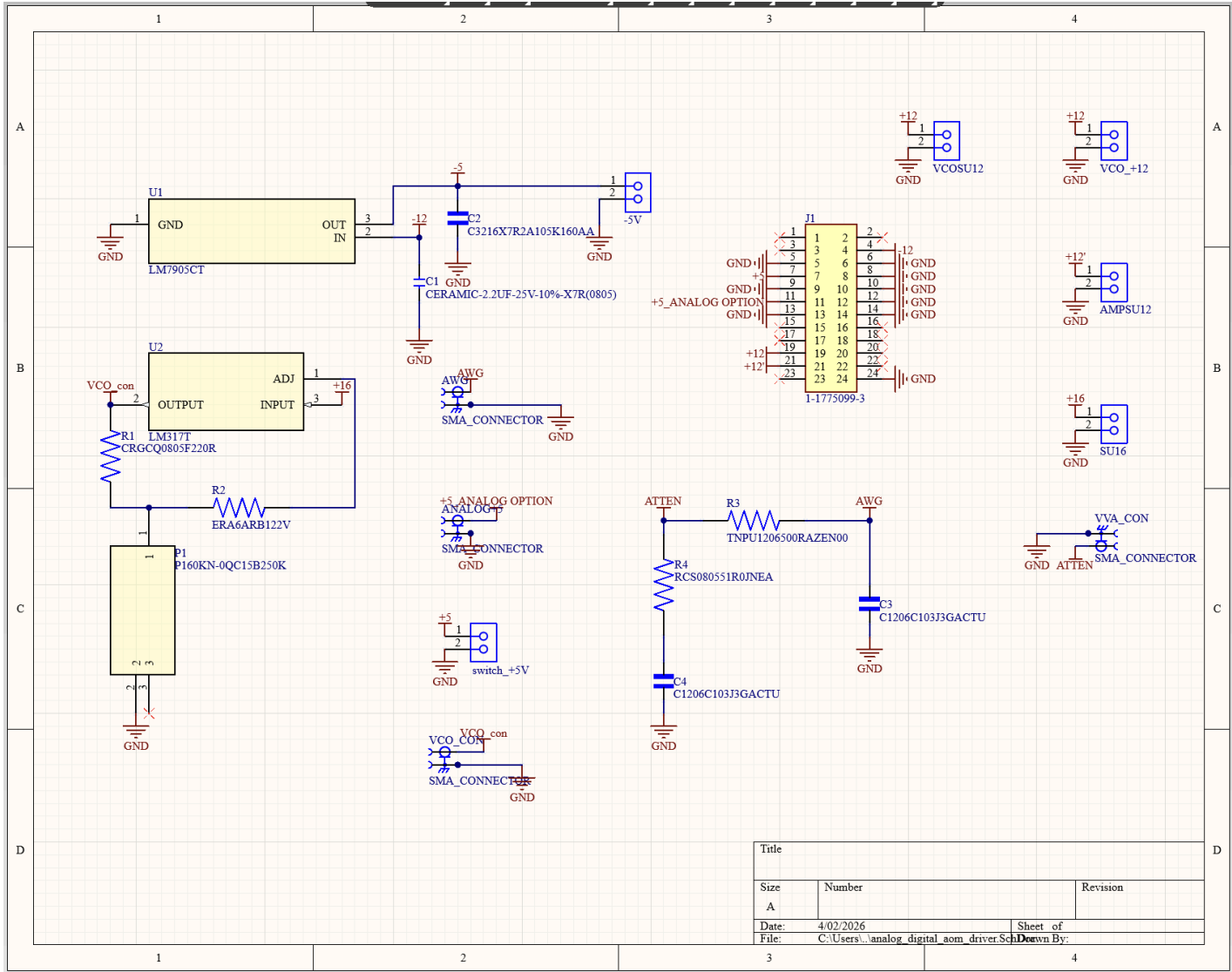
Figure 11: Final Enclosure



PCB Design

The final schematic was built in altium (Figure 12)

Figure 12: Altium Schematic



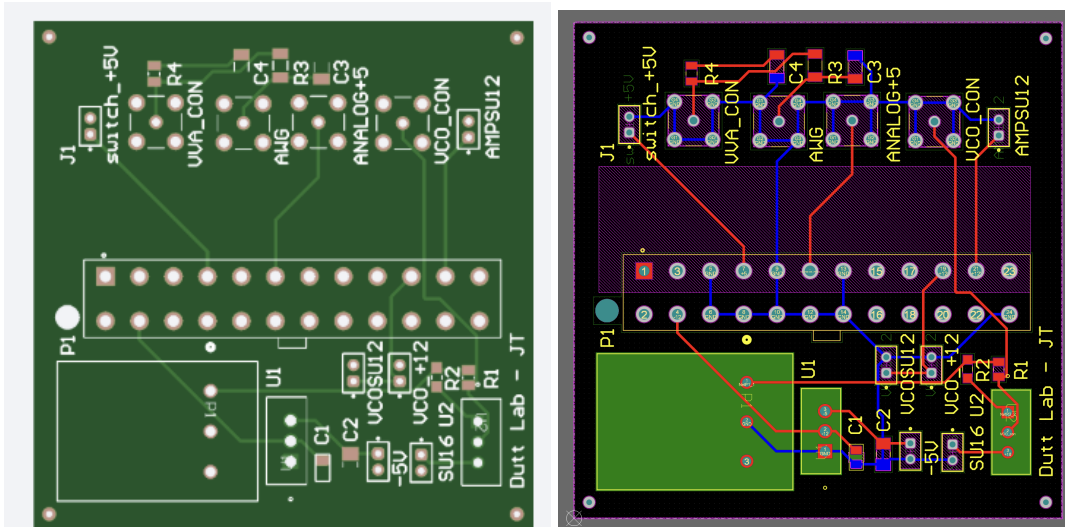
The PCB was built on Altium with 2-layer stackup, ground plane, and SMA connectors to ensure signal integrity

Surface-mount passives are used for noise reduction

JLCPCB design rule file passes

Exported GERBER files are shown in Figure 13

Figure 13: Analog and Digital AOM driver PCB



The final design evolved from a breadboard proof-of-concept to a compact PCB after verifying each subcircuit individually. The decision to use a computer PSU came after linear regulators proved insufficient for heat dissipation.

Design Testing

Test Plan

1. Power rail verification ($\pm 12V$, 5V, -5V (through 7905), 16V (through step up converter), 24V(through step up converter))
2. VCO frequency sweep (1–16V control $\rightarrow$ 150–300 MHz)
3. VVA attenuation control
4. TTL switching response
5. Combined analog + digital modulation
6. Final AOM diffraction test with laser pointer
7. PCB Test for Full Circuit

Test Results

The following tests passed for both the breadboard and PCB circuits

- Power rails: all within $\pm 5\%$ tolerance (used voltmeter)
- VCO: varied from 150 MHz to 300 MHz. Optimized at 210 MHz (crystal resonance) (used oscilloscope)
- TTL control works by connecting TTL from a function generator to TTL switch and sending a sine wave. The sine wave turns on an off corresponding to the TTL pulse (used oscilloscope)
- VVA attenuation control works as laser power changes by changing amplitudes from AWG (used Oscilloscope)
- TTL/VVA combined worked by applying TTL to the Switch and varying the AWG amplitudes on the VVA (used Oscilloscope)
- Max allowed RF output is 20 Vpp into 50Ω load. Our circuit provides 5.8Vpp on an oscilloscope with a 20dB attenuator $\rightarrow$ real circuit output is 58 Vpp (exceeds 1 W limit, therefore, the attenuator is kept) (used Oscilloscope)

- Diffraction observed at 210 MHz (see video demonstration) (used Photodiode and oscilloscope)

Debugging

Problem #1:

The amplifier required more current than stated in the datasheet → problem solved by using a higher power step up converter.

Summary, Conclusions, and Future Work

A functional hybrid analog/digital AOM driver was built from salvaged components, with added frequency control and AWG programmability. The final PCB improved noise performance and is fully functional.

The design met all primary goals: tunable frequency, hybrid analog/digital control, AWG programmability, and successful optical diffraction testing.

Future work

- Integrate step-up converters directly on PCB: This would eliminate external power modules, reduce wiring inductance, and improve reliability for field deployment
- Add pre-amplifier stage to reduce required input power
- Replace expensive Mini-Circuits modules with cheaper alternatives
- Investigate amplifier current draw discrepancy

REFERENCES:

- [1] VVA: ZAS-1: <https://www.minicircuits.com/pdfs/ZAS-1.pdf>
- [2] AMP: zhl-1-2w-s: <https://static.datasheets.com/doc/12201489-mini-circuits-zhl-1-2w-s-ds.pdf>
- [3] VCO: ZOS-300+: <https://www.minicircuits.com/pdfs/ZOS-300+.pdf>
- [4] TTL Switch: ZYSWA-2-50 DR: <https://www.alldatasheet.com/datasheet-pdf/view/236621/MINI/ZYSWA-2-50DR.html>
- [5] Variable Voltage regulator: LM317: <https://www.ti.com/product/LM317>
- [6] 7905 Voltage regulator: <https://rocelec.widen.net/view/pdf/mcm4dqve8e/lm79.pdf?t.download=true&u=5oefqw>
- [7] Proteus 1284M <https://www.taborelec.com/p1284m>
- [8] Keysight PXI Chassis <https://www.keysight.com/us/en/lib/software-detail/driver/pxie-chassis-family-drivers-2023724.html>
- [9] AOM <https://sphotonics.ru/upload/iblock/fa4/fa44349121a3f44939a023b2bd202fcf.pdf>
- [10] Photodiode <https://www.thorlabs.com/item/APD130A2>
- [11] Computer Power Supply <https://www.amazon.com/dp/B0FNHFXH5K>
- [12] Manufacturer-Provided Python Example Programs: <https://github.com/Tabor-Electronics/Python-Examples>